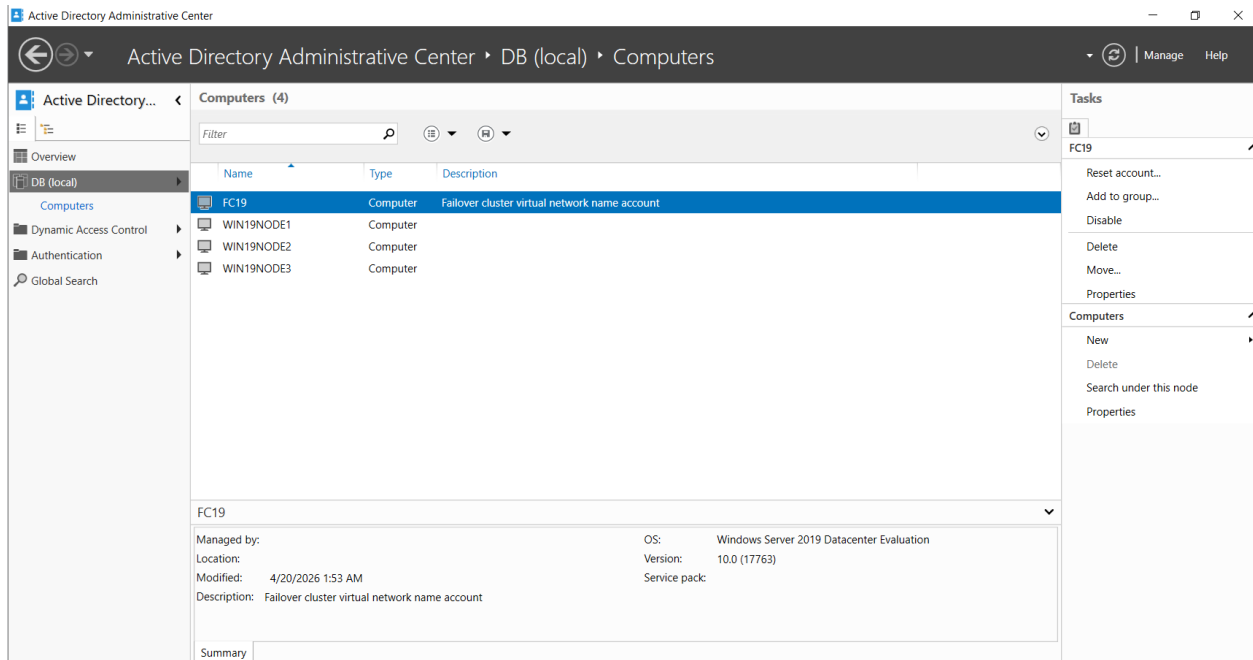


Implementation Steps

Step 1: Active Directory Setup

- Domain Controller configured (DB)
- All nodes joined to domain

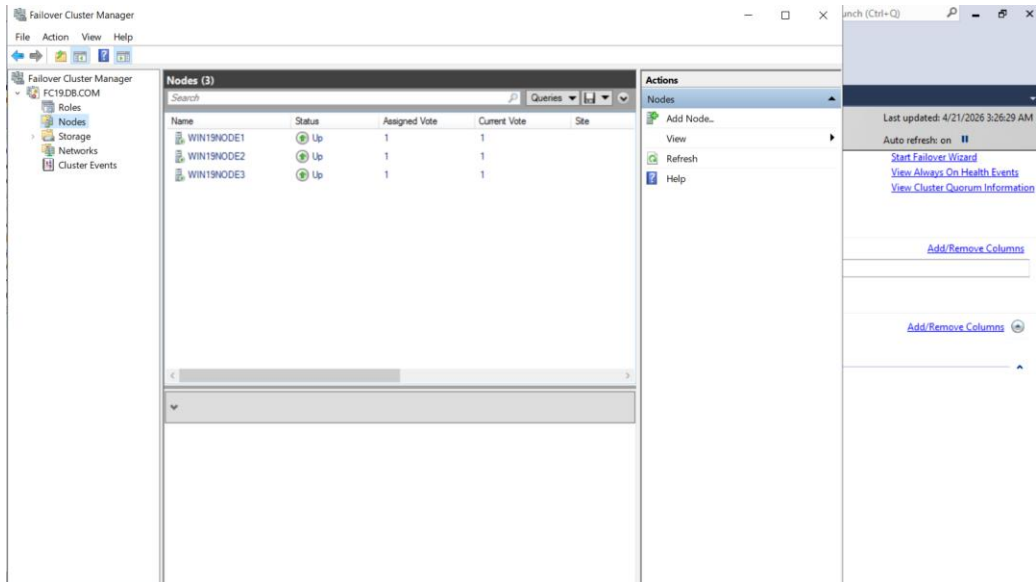


Step 2: Windows Failover Cluster (WSFC)

- Installed Failover Clustering feature
- Validation tests executed
- Cluster created

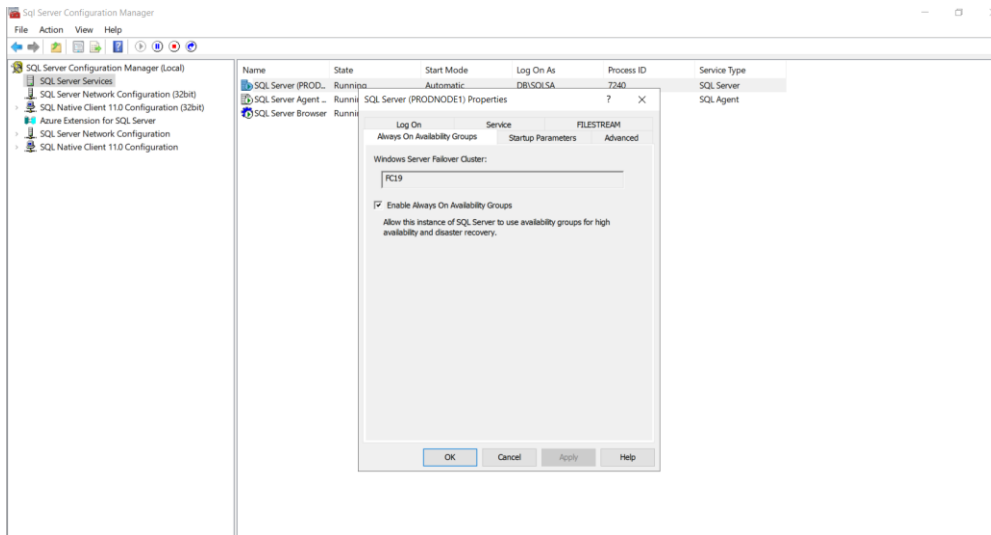
Cluster Name: FC19

Nodes: WIN19Node1, WIN19Node2, WIN19Node3



Step 3: SQL Server Configuration

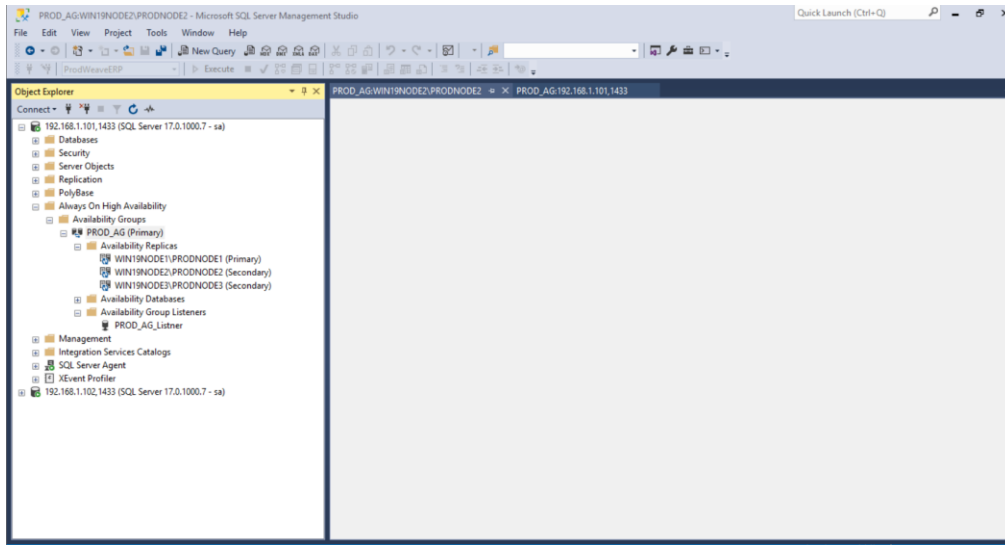
- Installed SQL Server on all nodes
- Enabled Always On feature



Step 4: Database Preparation

- Full backup
- Transaction log backup

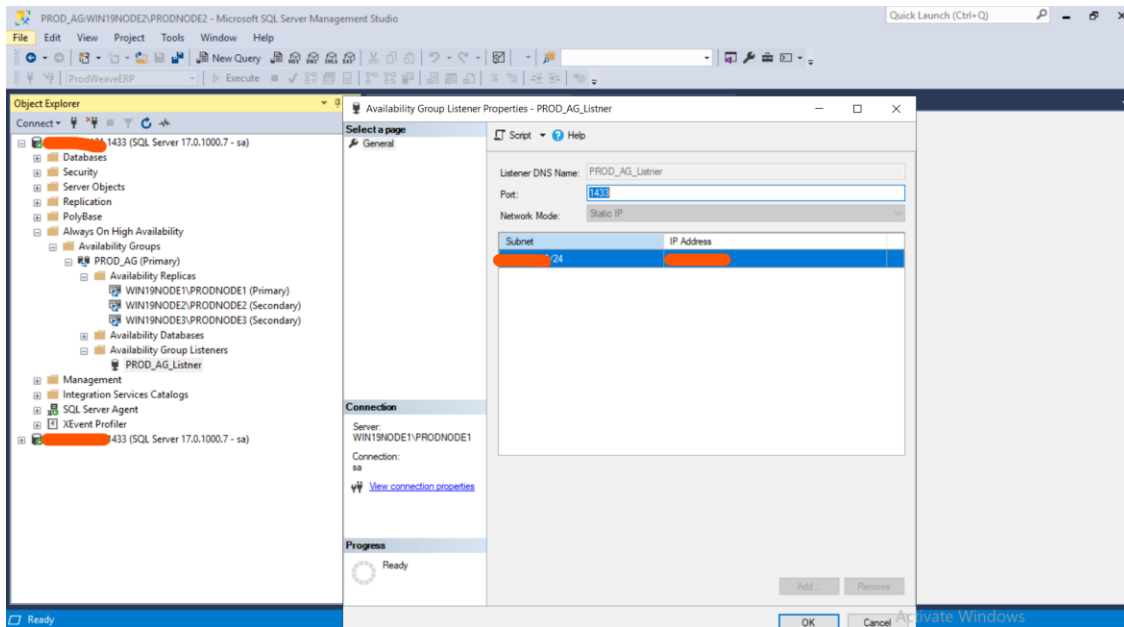
Step 5: Availability Group Creation



Step 6: Listener Configuration

Listener Name: PROD_AG_Listner

Port: 1433



Data Flow

Application → AG Listener → Primary Node

Primary Node:

Write Operations

↓ (Transaction Log)

Secondary Node 2:

Synchronous Replica (Real-time)

Secondary Node 3:

Asynchronous Replica (DR Site)

PROD_AG:192.168.1.101,1433 - Microsoft SQL Server Management Studio

File Edit View Project Tools Window Help

ProdWeaveERP

PROD_AG: hosted by WIN19NODE1\PRODNODE1 (Replica role: Primary)

Availability group state: Healthy

Primary instance: WIN19NODE1\PRODNODE1

Failover mode: Automatic

Cluster state: FC19 (Normal Quorum)

Cluster type: Windows Server Failover Cluster

Availability replica:

Name	Role	Availability Mode	Failover Mode	Seeding Mode	Synchronization State	Issues
WIN19NODE1\PRODNODE1	Primary	Synchronous commit	Automatic	Automatic	Synchronized	
WIN19NODE2\PRODNODE2	Secondary	Synchronous commit	Automatic	Automatic	Synchronized	
WIN19NODE3\PRODNODE3	Secondary	Synchronous commit	Automatic	Automatic	Synchronized	

Group by

Name	Replica	Synchronization State	Failover Read...	Issues
WIN19NODE1\PRODNODE1				
ProdWeaveERP	WIN19NODE1\PRODNODE1	Synchronized	No Data Loss	
WIN19NODE2\PRODNODE2				
ProdWeaveERP	WIN19NODE2\PRODNODE2	Synchronized	No Data Loss	
WIN19NODE3\PRODNODE3				
ProdWeaveERP	WIN19NODE3\PRODNODE3	Synchronized	No Data Loss	